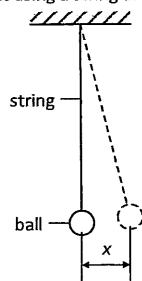


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

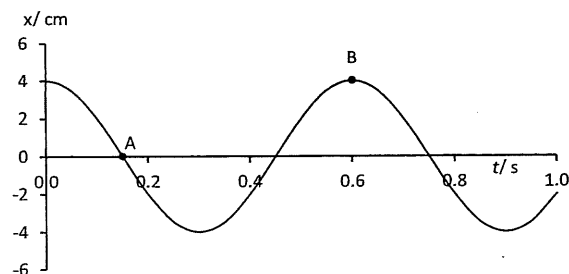
I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/13]

A small metal ball is suspended from a fixed point using a string as shown.



The ball is pulled a small distance to one side and then released. The variation with time t of the horizontal displacement x of the ball is shown below.



Which of the following statement is not true?

- A At A, the kinetic energy of the ball is maximum.
- B At A, the tension on the string is minimum.
- C At B, the ball is instantaneously at rest.
- D At B, the ball experiences the greatest restoring force.

2. [AJC/H2/Prelims 2018/P1/13]

An object of mass 40 g is oscillating with a frequency of 25 Hz.

What is the magnitude of the restoring force on the object when it is at a displacement of 2.5 mm from equilibrium?

- A 16 N B 9.8 N C 4.0 N D 2.5 N

3. [CJC/H2/Prelims 2018/P1/12], [JJC/H2/Prelims 2018/P1/19]

Fig. A shows the variation with displacement x of the velocity v of a body in simple harmonic motion. Fig. B shows the variation with time t of the net force F acting on the body.

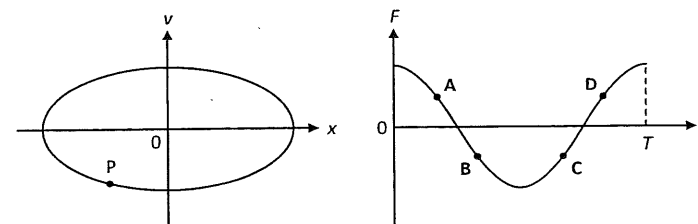


Fig. A

Fig. B

Which of the points in Fig. B corresponds to the state of motion represented by point P in Fig. A?

4. [DHS/H2/Prelims 2018/P1/14]

A particle performs simple harmonic motion according to the equation

$$x = 2.0 \cos(\omega t)$$

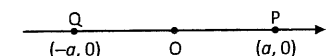
where its displacement x is measured in cm and time t is measured in s.

If the angular frequency $\omega = \pi \text{ rad s}^{-1}$, what is the total distance travelled by the particle from $t = 0.0 \text{ s}$ to $t = 1.5 \text{ s}$?

- A 0 cm B 2.0 cm C 3.0 cm D 6.0 cm

5. [DHS/H2/Prelims 2018/P1/15 Modified]

An object is executing simple harmonic motion along the x -axis between P($a, 0$) and Q($-a, 0$) about the origin O. The kinetic energy of the particle is E_k , its potential energy is E_p and the total energy is E_T .



When the particle is mid-way between O and Q, what are the values of E_k/E_T and E_p/E_T ?

	E_k/E_T	E_p/E_T
A	1/8	7/8
B	1/4	3/4
C	1/2	1/2
D	3/4	1/4

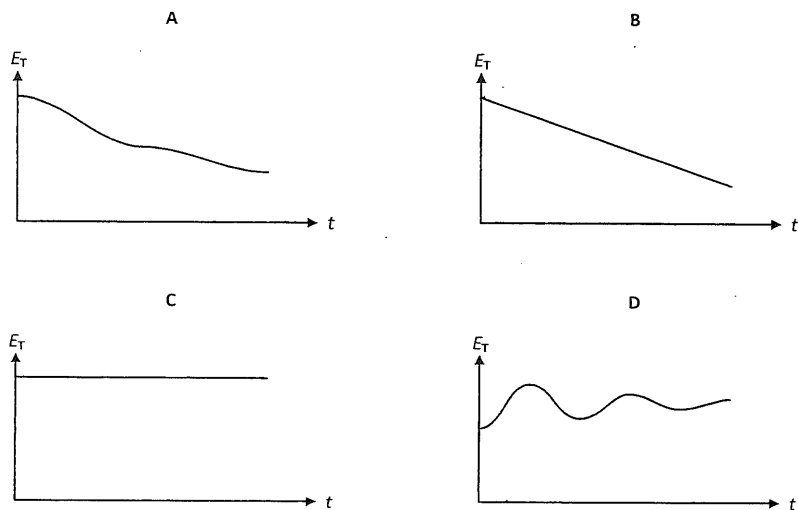
6. [EJC/H2/Prelims 2018/P1/12]

Which of the following statements is false in describing a mass moving in simple harmonic motion?

- A The maximum kinetic energy is dependent on the frequency of the oscillation.
- B The time taken for the system to change from maximum kinetic energy to maximum potential energy is a quarter of the period of the oscillation.
- C An oscillating system with larger amplitude will have a longer period.
- D An oscillating system with larger amplitude will have a greater maximum velocity.

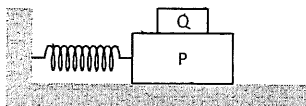
7. [EJC/H2/Prelims 2018/P1/13 Modified]

Which of the following graphs show the variation in the total energy E_T of an object under light damping as time t passes?



8. [HCl/H2/Prelims 2018/P1/13]

A system consisting of a large block P with a smaller block Q resting on it, oscillates on a frictionless surface with a frequency of 1.5 Hz. The maximum static friction between the two blocks is 5.0 N.



If the mass of P is 2.00 kg and the mass of Q is 0.20 kg, what is the maximum amplitude of oscillation of the system in order that block Q does not slip?

- A 0.28 m B 0.056 m C 0.028 m D 0.026 m

9. [UC/H2/Prelims 2018/P1/14]

A particle oscillates with undamped simple harmonic motion.

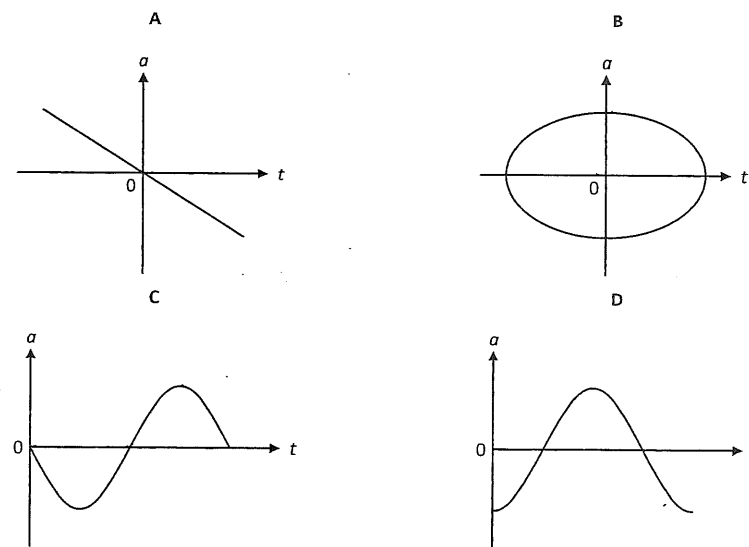
Which one of the following statements about the acceleration of the oscillating particle is true?

- A It decreases as the potential energy increases.
 B It is always in the opposite direction to its velocity.
 C It is least when the speed is greatest.
 D It is proportional to the frequency.

10. [MI/H2/Prelims 2018/P1/15]

The bob of a pendulum is moving in simple harmonic motion. The bob has maximum potential energy at time $t = 0$ s.

Which of the following graphs best describes the variation of the acceleration a of the bob with time t ?



11. [MI/H2/Prelims 2018/P1/16]

A mass of 20 g is oscillating vertically in simple harmonic motion. The displacement of the mass is given by the equation

$$x = 6.0 \times 10^{-3} \sin(3\pi t)$$

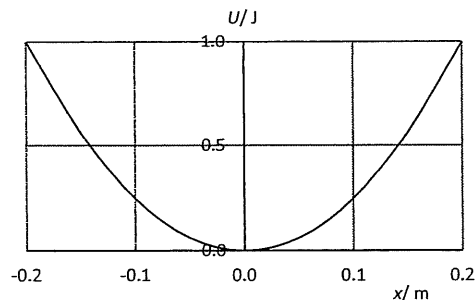
where x is in metres and t in seconds.

What is the magnitude of the maximum force acting on the mass?

- A 0.01 N B 0.50 N C 1.80 N D 10.7 N

12. [MJC/H2/Prelims 2018/P1/9]

A particle of mass 4.0 kg moves in simple harmonic motion. Its potential energy U varies with position x as shown in the figure below.



What is the period of oscillation of the mass?

- A $\frac{2\pi}{25}$ s B $\frac{2\pi\sqrt{2}}{5}$ s C $\frac{8\pi}{25}$ s D $\frac{4\pi}{5}$ s

13. [MJC/H2/Prelims 2018/P1/10]

A toy car moving along a horizontal plane in simple harmonic motion starts from the amplitude at time $t=0$ s.

If the amplitude of its motion is 5.0 cm and frequency is 2.0 Hz, the magnitude of the acceleration of the toy car at 1.7 s is

- A 0.25 m s^{-2} B 0.51 m s^{-2} C 6.4 m s^{-2} D 7.4 m s^{-2}

14. [NJC/H2/Prelims 2018/P1/11]

A student sets up a simple harmonic oscillator comprising a mass m and a light spring with force constant k which oscillates with a period T . At equilibrium, the extension of this spring is x_0 .

To overcome air resistance, the length of the spring is halved, and the oscillator is attached to an actuator (driver). At the new equilibrium point, the extension of this spring is now $\frac{1}{2}x_0$.

What frequency should the actuator be set so that the oscillator can achieve maximum amplitude?

- A $\frac{1}{T\sqrt{2}}$ B $\frac{\sqrt{2}}{T}$ C $\frac{1}{T}$ D $\frac{2}{T}$

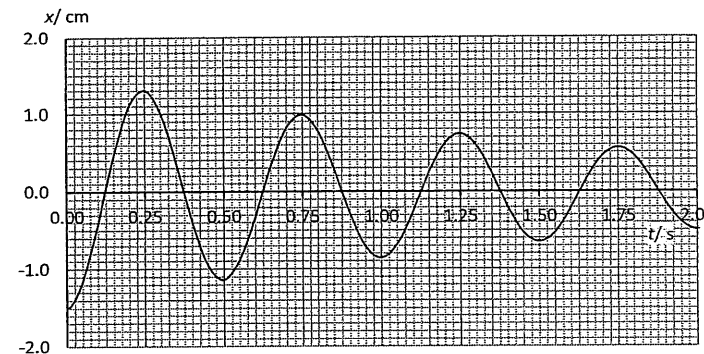
15. [NYJC/H2/Prelims 2018/P1/13]

Which of the statements about damped oscillations is not true?

- A The period of the oscillation increases with time.
B The amplitude of the oscillation decreases with time.
C The energy of the oscillation decreases with time.
D The frequency of the oscillation is lower compared to that of an undamped oscillation.

16. [PJC/H2/Prelims 2018/P1/8 Modified]

A wooden block of mass 200 g floats on water and oscillates vertically. The variation with time t of its vertical displacement x is as shown.



What is the decrease in maximum potential energy of the oscillations during the first 1.5 s?

- A 0.7×10^{-3} J B 2.9×10^{-3} J C 3.6×10^{-3} J D 1.7×10^{-2} J

17. [RVHS/H2/Prelims 2018/P1/13]

A particle executes simple harmonic motion about a point. Its maximum speed is 6.0 m s^{-1} and its maximum acceleration is 2.0 m s^{-2} .

What is the amplitude and the period of oscillations?

	amplitude/ m	period/ s
A	0.33	0.35
B	0.33	18
C	18	0.33
D	18	19

18. [RVHS/H2/Prelims 2018/P1/14 Modified]

A system oscillates freely with a period of 1.0 s. A periodic force of 2.0 Hz is then applied to the system, reaching a steady state.

What is the time interval between successive instances where the magnitude of the displacement is at a maximum?

- A 0.25 s B 0.50 s C 0.75 s D 1.0 s

19. [TPJC/H2/Prelims 2018/P1/17]

A body performs simple harmonic motion with a period of 0.063 s. The maximum speed of the body is 3.0 m s^{-1} . What is the amplitude x_0 and the angular frequency ω of oscillations?

	x_0 / m	$\omega / \text{rad s}^{-1}$
A	0.030	100
B	0.19	16
C	5.3	16
D	33	100

20. [VJC/H2/Prelims 2018/P1/12]

A particle is oscillating with simple harmonic motion described by the equation

$$y = 5 \sin(20\pi t)$$

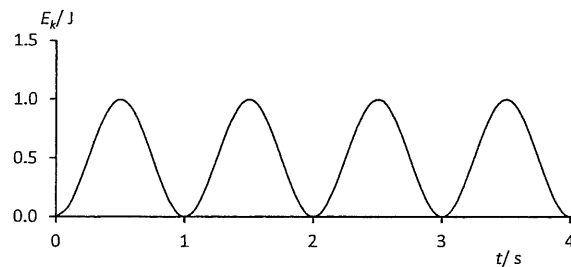
where y is in meters and t is in seconds.

How long does it take the particle to travel from its position of maximum displacement to a position midway between the maximum displacement and the equilibrium position?

- A Less than 12.5 ms B 12.5 ms
C Between 12.5 ms and 25 ms D 25 ms

21. [VJC/H2/Prelims 2018/P1/13]

The variation with time t of the kinetic energy E_k of a particle undergoing simple harmonic motion with amplitude of 3.0 cm is as shown.



What is the maximum velocity of the particle?

- A 0.59 m s^{-1} B 0.30 m s^{-1} C 0.19 m s^{-1} D 0.094 m s^{-1}

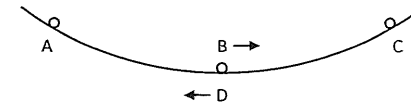
22. [YJC/H2/Prelims 2018/P1/13]

Which one of the following statements always applies to a damping force acting on an oscillatory system?

- A It is the same direction as the acceleration.
B It is the opposite direction to the velocity.
C It is the same direction as the displacement.
D It is proportional to the displacement.

23. [YJC/H2/Prelims 2018/P1/14]

A ball bearing rolls on a concave surface in approximately simple harmonic motion. It is released at A and passes through the lowest point B before reaching C. It then returns through the lowest point D.



At which stage, A, B, C or D, does the ball bearing experience maximum acceleration to the left?

II. Structured Questions (ACJC –YJC)

1. [ACJC/H2/Prelims 2018/P3/4]

A mass damper can be used to stabilize a building during earthquakes. A mass-spring system shown in Fig 1.1 can be used to model a mass damper.

A 600 g mass is placed on a smooth surface and is attached horizontally to two unstretched identical springs X and Y with a spring constant of 20 N m^{-1} .

When the mass is displaced by 5.0 cm from its equilibrium position and released from rest as shown in Fig. 1.2, it undergoes simple harmonic motion.

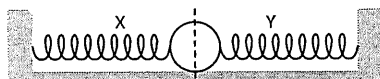


Fig. 1.1

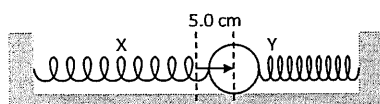


Fig. 1.2

- a. i. Using energy consideration, calculate the maximum velocity of the mass. [3]
- ii. Hence, determine the period of the oscillating spring-mass system. [2]
- iii. Sketch on Fig 1.3 the variation with time t from the point of release of the mass
 1. of the kinetic energy of the mass. Label this graph M. [2]
 2. of the elastic potential energy of spring Y. Label this graph S.

Show 1 complete oscillation.

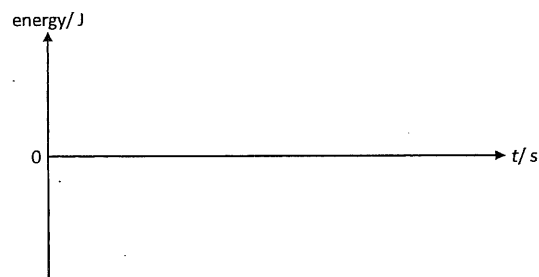


Fig. 1.3

- b. Suggest how the mass damper can help stabilize a building during an earthquake. [1]

[Total: 8]

2. [AJC/H2/Prelims 2018/P2/4 Modified]

- a. State, for an oscillating system, what is meant by
 - i. natural frequency of vibration, [1]
 - ii. resonance. [2]
- b. State and explain one situation where resonance is useful. [2]
- c. A car component of mass 0.430 kg undergoes forced oscillation. Fig. 4.1 shows how the amplitude A of the oscillation varies with frequency f .

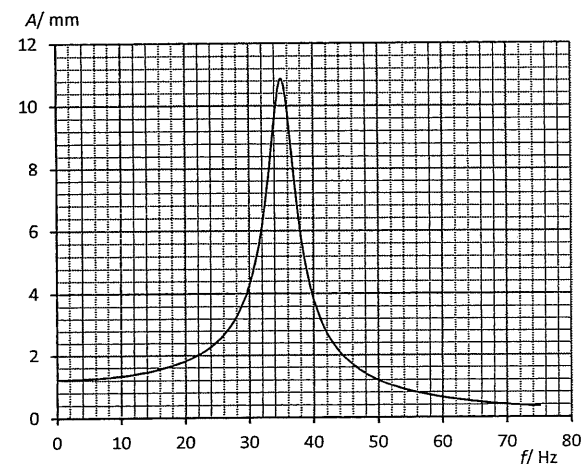


Fig. 2.1

- i. Calculate, for the component oscillating at the resonant frequency, the time interval between a maximum linear speed and the subsequent maximum linear acceleration. [3]
- ii. The component is now supported on a rubber mounting. On Fig. 2.1, draw a line to show the variation with frequency of amplitude for the supported component. [2]

[Total: 10]

3. [CJC/H2/Prelims 2018/P3/3 Modified]

The variation with displacement x of the acceleration a of an animal's eardrum is shown in Fig. 3.1.

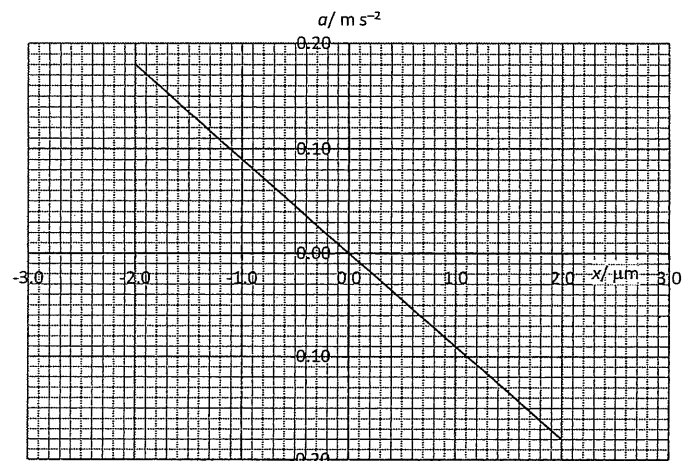


Fig. 3.1

- Explain how Fig. 3.1 shows that the motion of the eardrum is simple harmonic. [2]
- Calculate the time taken for the eardrum to travel a distance of $0.50 \mu\text{m}$ starting from its maximum displacement towards the equilibrium point. [3]
- The mass of the eardrum is 14 mg .
Calculate the potential energy of the eardrum when its displacement is $0.75 \mu\text{m}$. [2]

[Total: 7]

4. [DHS/H2/Prelims 2018/P2/3 Modified]

- A small ball rests at point P on a curved track of radius r , as shown in Fig. 4.1.

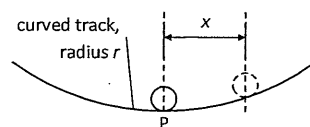


Fig. 4.1

The ball is moved a small distance to one side and is then released. The horizontal displacement x of the ball is related to its acceleration a towards P by the expression

$$a = -\frac{gx}{r}$$

where g is the acceleration of free fall.

- Show that the ball undergoes simple harmonic motion. [2]
- The radius r of curvature of the track is 28 cm .
Determine the time interval τ between the ball passing point P and then returning to point P. [3]

- The variation with time t of the displacement x of the ball in (a) is shown in Fig. 4.2.

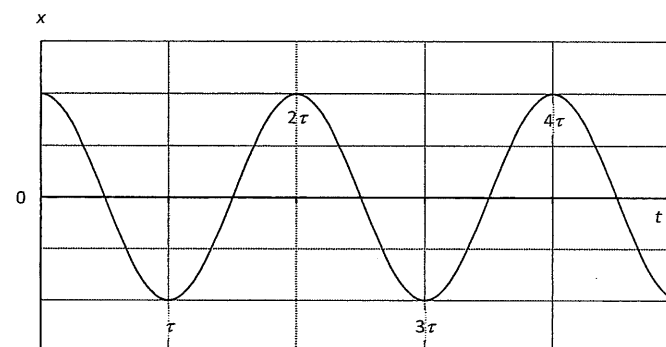


Fig. 4.2

Some moisture now forms on the track, causing the ball to come to rest after approximately 15 oscillations.

On the axes of Fig. 4.2, sketch the variation with time t of the displacement x of the ball for the first two periods after the moisture has formed. Assume the moisture forms at $t=0$. [3]

[Total: 8]

5. [EJC/H2/Prelims 2018/P3/2]

- State what is meant by *simple harmonic motion*. [2]
- A small frictionless trolley is attached to a fixed point P by means of a spring. A second spring is used to attach the trolley to a variable frequency oscillator, as shown in Fig. 5.1.

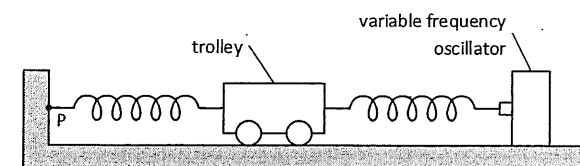


Fig. 5.1

Both springs remain extended within their limits of proportionality. Initially, the oscillator is switched off. The trolley is displaced horizontally along the line joining the two springs and is then released. The variation with time t of the velocity v of the trolley is shown in Fig. 5.2.

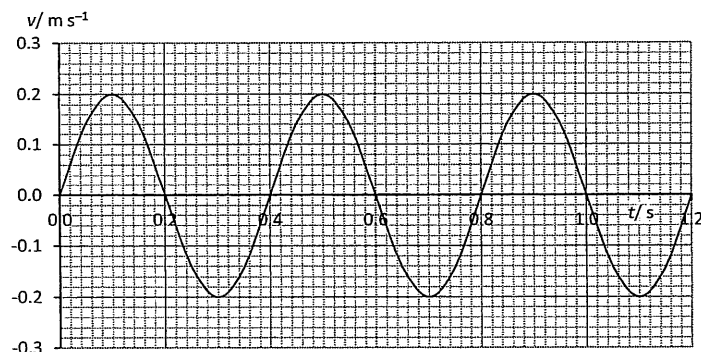


Fig. 5.2

- i. Using Fig. 5.2, state two different times at which
 1. the displacement of the trolley is zero, [1]
 2. the acceleration in one direction is maximum. [1]
- ii. Determine the frequency of oscillation of the trolley. [1]
- iii. The variation with time of the displacement of the trolley is sinusoidal. The variation with time of the velocity of the trolley is also sinusoidal.
State the phase difference between the displacement and the velocity. [1]
- c. The oscillator is now switched on. The amplitude of variation of the oscillator is constant. The frequency f of vibration of the oscillator is varied. The trolley is forced to oscillate by means of vibrations of the oscillator.
 - i. Distinguish between *free oscillations* and *forced oscillations*. [2]
 - ii. It was observed that the trolley vibrates with different amplitudes as the frequencies of the oscillator changes.
On the axis provided on Fig. 5.3, sketch a possible amplitude-driving frequency graph for this trolley. [1]

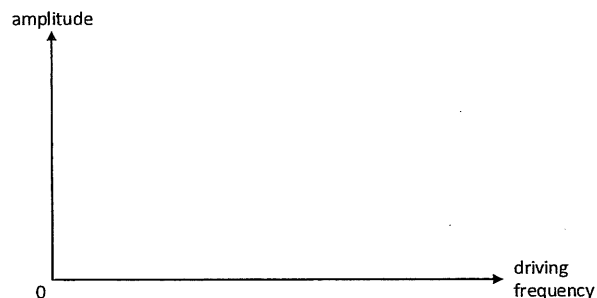


Fig. 5.3

- iii. State the approximate frequency at which the amplitude is maximum. [1]
- iv. The amplitude of the oscillations may be reduced without changing significantly the frequency at which the amplitude is a maximum.
State and explain how this may be done. You may draw on Fig. 5.1 to support your answer. [2]

[Total: 12]

6. [HCI/H2/Prelims 2018/P2/1]

- a. A spring, which has an unstretched length of 0.650 m, is attached to a fixed point. A mass of 0.400 kg is attached to the spring and gently lowered until equilibrium is reached. The spring has then stretched elastically by a distance of 0.200 m.

Calculate, for the stretching of the spring,

- i. the loss in gravitational potential energy of the mass, [1]
- ii. the elastic potential energy gained by the spring. [2]

- b. Explain why the two answers to a. are different. [2]

- c. The mass on the spring is now set into simple harmonic motion by pulling it downwards by an additional 0.100 m and released from rest.

The angular frequency ω of a spring-mass system is given by $\omega = \sqrt{k/m}$, where k is the force constant and m is the mass of the load.

- i. Calculate the maximum speed of the mass. [2]
- ii. The velocity-displacement graph of the mass is shown in Fig. 6.1. Label on the graph the point P when the mass is first released and point Q when it first returns to its equilibrium position. Include numerical values on the axes. Take upwards to be positive. [1]

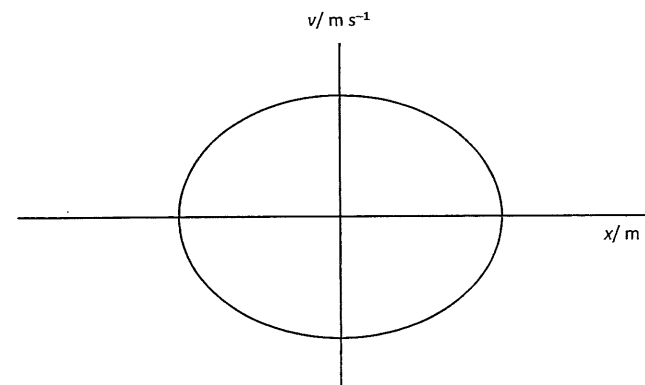


Fig. 6.1

- iii. The mass is now lowered into a container of water until it is completely submerged. The mass is again displaced downwards from its equilibrium position by 0.100 m and released from rest. Throughout the oscillations, the mass remains under water.
Sketch on Fig. 6.1 the variation of velocity with respect to displacement for one complete oscillation of the mass. [2]

[Total: 10]

7. [JJC/H2/Prelims 2018/P3/4]

- a. Distinguish between free oscillations and forced oscillations. [2]
- b. A trolley is held on a horizontal surface by means of two stretched springs, as shown in Fig. 7.1.

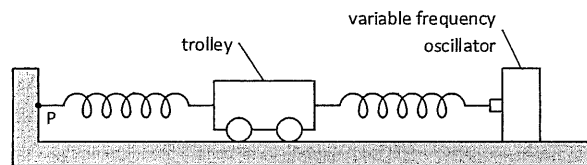


Fig. 7.1

One spring is attached to a fixed point P. The other spring is attached to an oscillator that causes horizontal oscillations of the trolley.

The oscillator vibrates with a constant amplitude of vibration. The frequency of vibration of the oscillator is gradually increased from a very low value.

The variation with frequency f of the amplitude x_0 of vibration of the trolley is shown in Fig. 7.2.

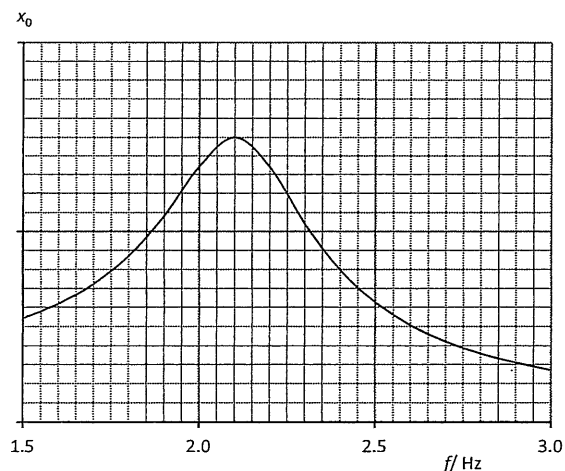


Fig. 7.2

Use Fig. 7.2 to state and explain

- i. the value of the natural frequency of vibration of the trolley, [3]
- ii. whether there are any frictional forces acting on the trolley. [1]
- c. The oscillator in b. is now stopped. The trolley is given a horizontal displacement of 4.7 cm along the line of the springs. The trolley is then released. Use information from Fig. 7.2 to estimate the maximum speed of the trolley. [2]

[Total: 8]

8. [JJC/H2/Prelims 2018/P2/4 Modified]

The needle of a sewing machine is made to oscillate vertically through a distance of 22 mm.

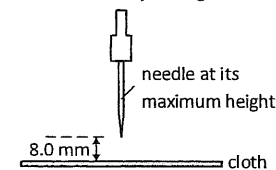


Fig. 8.1

The oscillations of the needle are simple harmonic with a frequency of 4.5 Hz. The cloth that is being sewn is positioned 8.0 mm below the point of the needle when the needle is at its maximum height as shown in Fig. 8.1.

- a. Explain what is meant by *simple harmonic motion*. [2]
- b. Determine the values of
- the amplitude of oscillation, [1]
 - the angular frequency, [1]
 - the maximum acceleration, [1]
 - the displacement of the needle from its equilibrium position as it just passes downwards through the cloth (taking upward direction as positive), [1]
 - the speed of the needle as it just passes downwards through the cloth. [2]

[Total: 8]

9. [MI/H2/Prelims 2018/P2/3 Modified]

- a. State what is meant by *simple harmonic motion*. [2]
- b. A smooth ball of mass m is held between two fixed points A and B by two identical springs, each of spring constant k , as shown in Fig. 9.1.

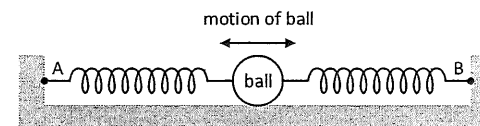


Fig. 9.1

When the ball is in equilibrium, the extension of each spring is e . The ball is then displaced a small distance x to the right along the axis of the springs. The ball is then released and oscillates on the smooth surface along the straight line joining points A and B.

- i. Show that the acceleration a of the ball is given by the equation

$$a = -\frac{2kx}{m}$$

[3]

- ii. The mass m of the ball is 900 g and the spring constant k is 120 N m⁻¹. The maximum acceleration of the ball is 5.2 m s⁻². Determine
- the frequency of oscillation of the ball, [2]
 - the amplitude of the oscillation, [2]
 - the maximum kinetic energy of the ball. [2]

[Total: 11]

10. [MJC/H2/Prelims 2018/P2/1]

a. For an oscillating body, state what is meant by

- natural frequency of vibration,
- resonance.

[1]

[1]

b. State and explain one situation where resonance is useful.

[2]

c. In some situations, resonance should be avoided.

State one such situation and how the effects of resonance are reduced.

[2]

[Total: 6]

11. [NJC/H2/Prelims 2018/P2/3 Modified]

A boy fills a flask with water and places it on a mass balance. The boy records the reading on the mass balance x in grams. He inserts two similar cubes, one on top of the other, into the flask as shown in Fig. 11.1.

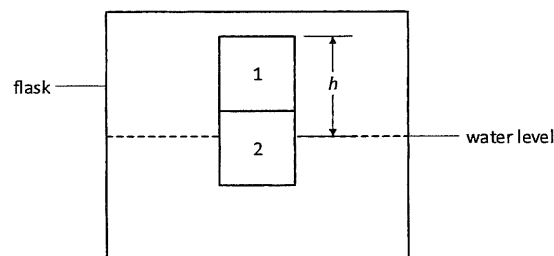


Fig. 11.1

Each side of the cube is r in cm. Both cubes stay afloat with one cube being partially immersed. After some time, the boy records that top of the cube 1 is a height h (in cm) above the water level and the reading on the mass balance is now y in grams.

a. Express the density of the fluid in terms of x , y , r and h in g cm^{-3} .

[2]

b. Cube 1 is removed. Suggest and explain the subsequent motion of cube 2.

[4]

c. Given that $r = 2.0$ cm and $h = 3.2$ cm, calculate the height of the top of cube 2 from the water level at equilibrium.

[2]

d. Sketch the variation with time t of

i. the height h of the top of cube 2 above water level,

[2]

ii. the kinetic energy E_k of cube 2.

[2]

Note: The top of cube 2 is 1.2 cm above the water level before cube 1 is removed.

[Total: 12]

12. [NYJC/H2/Prelims 2018/P2/3]

A peg is fixed to the rim of a vertical turntable of radius r , rotating with a constant angular speed ω , as shown in Fig. 12.1.

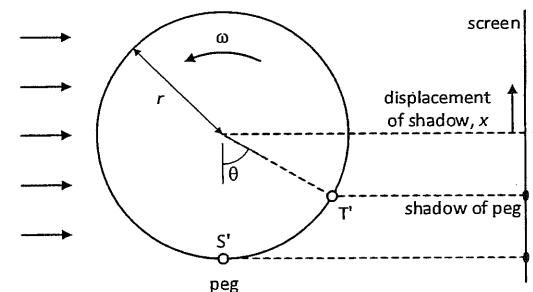


Fig. 12.1

A parallel beam of light is incident on the turntable such that the shadow of the peg is observed on the screen. Initially, the peg is at position S' and its shadow is at S . After time t , the peg moves through an angle of θ and it is positioned at T' while its shadow is at T . The displacement of the shadow, x , from O is shown in Fig. 12.1 where the upward direction is taken to be positive.

a. i. Express the angular displacement of the peg, θ , in terms of ω and t .

[1]

ii. Write down an expression for the displacement, x , of the shadow on the screen in terms of ω , t and r .

[1]

iii. Hence, prove that the shadow of the peg is moving in simple harmonic motion. Explain your working.

[2]

b. The turntable has a radius of 20 cm and angular speed of 3.5 rad s^{-1} . For the motion of the shadow on the screen,

i. calculate the magnitude of the acceleration of the shadow when the shadow is instantaneously at rest,

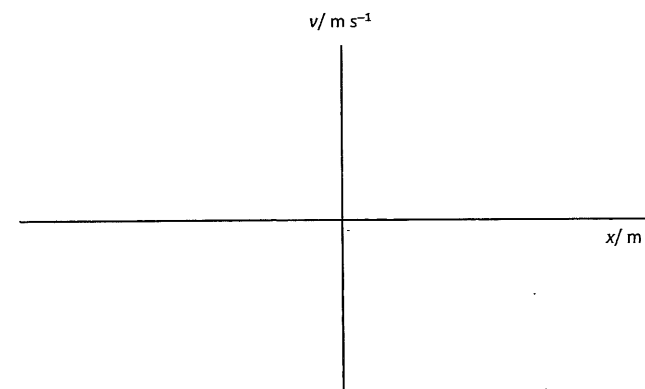
[1]

ii. state the velocity of the shadow as it passes through S ,

[1]

iii. sketch the variation with its displacement x of the velocity v of the shadow.

[2]



[Total: 8]

13. [PJC/H2/Prelims 2018/P3/9 Modified]

- a. Distinguish between *velocity* and *angular velocity*. [2]
- b. A toy pendulum consists of a sphere of mass 0.800 kg, attached firmly to one end of a rod of negligible mass. The other end of the rod is pivoted freely at point O. The distance between the centre of gravity of the toy to O is 0.600 m. The structure is held in a position such that the rod is at an angle of 6.0° from the vertical, as shown in Fig. 13.1.

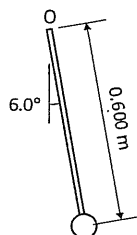


Fig. 13.1

The structure is then released from rest and oscillates in simple harmonic motion. At one instant during the oscillation, the toy is directly below O, as shown in Fig. 13.2.

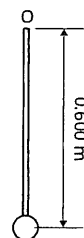


Fig. 13.2

- i. On Fig. 13.2, indicate the forces acting on the sphere. [2]
- ii. Show that the linear speed of the sphere at the instant in Fig. 13.2 is 0.254 m s^{-1} . [2]
- iii. Determine the force F exerted on the pivot by the rod in Fig. 13.2. [2]
- iv. Sketch, with appropriate values, the variation with time t of the velocity v of the sphere assuming the sphere is at maximum displacement at $t = 0 \text{ s}$. Label the period T . [2]
- c. i. State what is meant by *resonance*. [2]
- ii. A ship of mass 1020 kg undergoes simple harmonic motion in the vertical direction due to surface water waves that are incident on the ship. The water waves of amplitude of 0.18 m and frequency 0.56 Hz cause resonance in the vertical motion of the ship.
- Write an expression for the vertical displacement y as a function of the time t associated with the motion of the ship. [2]
 - Calculate the magnitude of acceleration of the sailing boat when its displacement from its equilibrium position is 0.080 m. [2]
 - Calculate the maximum kinetic energy of the ship during the vertical oscillation. [2]
 - Sketch, with appropriate values, the variation with time t of the kinetic energy E_k of the ship. [2]

[Total: 20]

14. [RVHS/H2/Prelims 2018/P2/4 Modified]

Block P of mass 5.0 kg is resting on a frictionless surface and is in equilibrium while connected to a spring as shown in Fig. 14.1. The spring that is fastened to the wall has an elastic constant $k = 100 \text{ N m}^{-1}$.

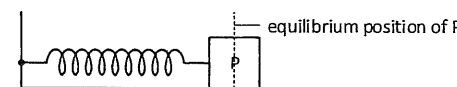


Fig. 14.1

- a. i. P is displaced to the right and then released.
Explain why the subsequent horizontal motion of P while connected to the spring will be simple harmonic. Draw a free-body diagram of P in the space below to illustrate your answer. [3]
- ii. Hence or otherwise, show that the period of oscillation for P is 1.4 s. [2]
- b. A second block Q of mass 3.0 kg is slowly pushed against P, compressing the spring by 0.20 m as shown in Fig. 14.2.

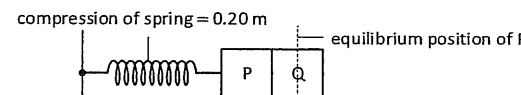


Fig. 14.2

The system is then released, and both blocks start moving to the right on the frictionless surface.

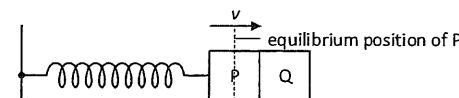


Fig. 14.3

- i. When P reaches the equilibrium point, Q loses contact with P and moves to the right with speed v . Determine speed v . [2]
- ii. Determine the amplitude of the oscillatory motion of P after it loses contact with Q. [1]
- iii. Hence, determine the gap between P and Q when the spring is fully stretched for the first time. [2]
- c. While P is oscillating about the equilibrium point, block R is dropped on it just when P is at its amplitude. R sticks to P and then moves along with P for the subsequent motion.

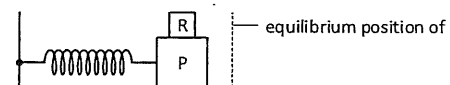


Fig. 14.4

Explain how the subsequent frequency and amplitude of the motion will be different. [2]

[Total: 12]

15. [SRJC/H2/Prelims 2018/P2/3 Modified]

- a. A block is placed on a smooth horizontal surface and is connected to a light spring, as shown in Fig. 15.1.

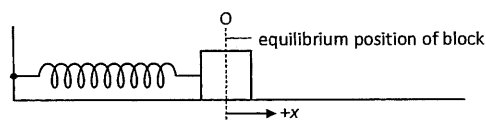


Fig. 15.1

The block is displaced to the right by 0.60 m and then released. Fig. 15.2 shows the variation with displacement x of acceleration a of the block.

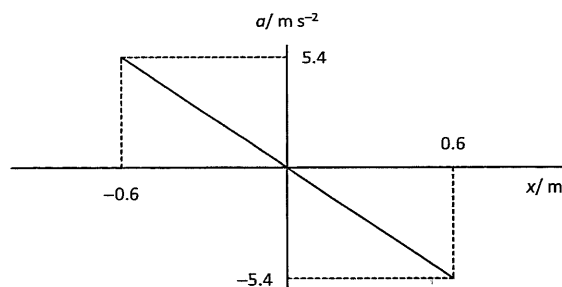


Fig. 15.2

- i. Use two features of the graph in Fig. 15.2 to explain why the motion of the block is simple harmonic. [2]
- ii. Calculate
 1. the angular frequency, [1]
 2. the maximum speed of the block. [1]
- iii. Sketch on Fig. 15.3, the variation with time t of the kinetic energy E_k of the block for one complete oscillation. The mass of the object is 0.0020 kg. Assume that the object just passes by the equilibrium position at time $t=0$ s. [2]

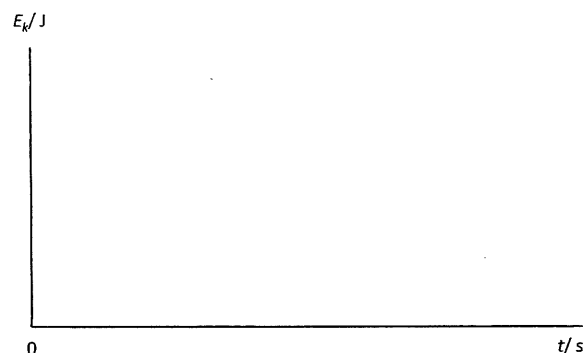


Fig. 15.3

- iv. Calculate the shortest time taken for the object to move from a point 0.30 m to the left of the equilibrium point to a point 0.30 m to the right of the equilibrium point. [3]
- b. A car component of mass 0.0460 kg rattles at a resonant frequency of 35.5 Hz. Fig. 15.4 shows the variation with driving frequency f of the amplitude x_0 of the vertical oscillation.

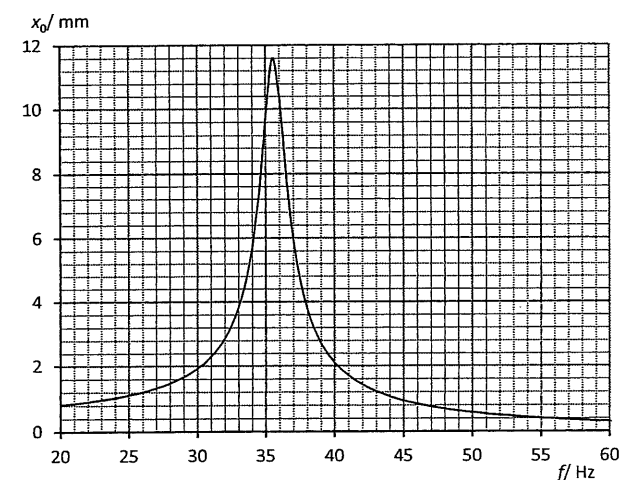


Fig. 15.4

- i. When oscillating at the resonant frequency, calculate
 1. the angular frequency of the oscillation, [1]
 2. the total energy stored in the oscillation of the component. [2]
- ii. Draw on Fig 15.4 to show how the amplitude of the oscillation varies with frequency if the component is supported on a rubber mounting. [2]

[Total: 14]

16. [TPJC/H2/Prelims 2018/P2/3]

- a. State what is meant by *simple harmonic motion*. [2]
- b. A spring hangs vertically from a fixed point. A copper plate is attached to the free end of the spring as illustrated in Fig. 16.1.

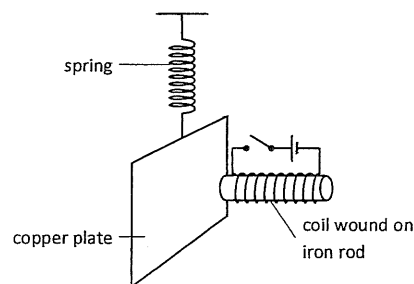


Fig. 16.1

One end of a coil of wire, wound on an iron rod, is placed near to the copper plate.

The copper plate is displaced vertically and then released. The variation with time t of the vertical displacement y of the plate is shown in Fig. 16.2.

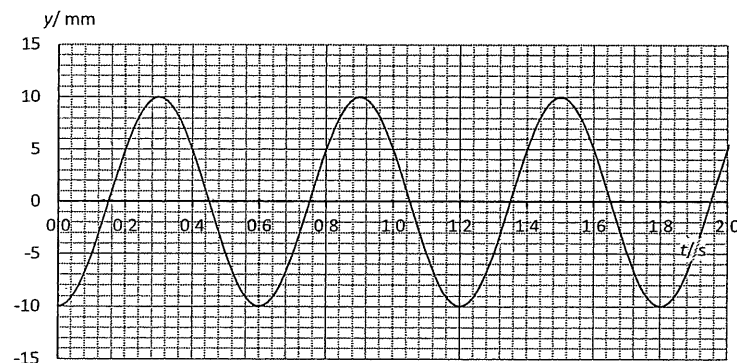


Fig. 16.2

The copper plate undergoes simple harmonic motion. The mass m of the oscillating copper plate is 320 g.

- i. Determine the frequency f of oscillation of the copper plate. [1]
- ii. Determine the maximum acceleration of the copper plate. [2]
- iii. Using your answer from ii., determine the maximum force exerted by the spring on the copper plate. [2]
- c. When the switch is closed, the oscillation of the copper plate is lightly damped.
- i. On Fig 16.2, sketch the variation with time t of the vertical displacement y of the plate when the switch is closed. [2]
- ii. State the energy transformations that occur in the copper plate when the switch is closed. [1]

[Total: 10]

17. [VJC/H2/Prelims 2018/P2/3 Modified]

- a. Fig. 17.1 shows a rectangular block of wood floating in water.

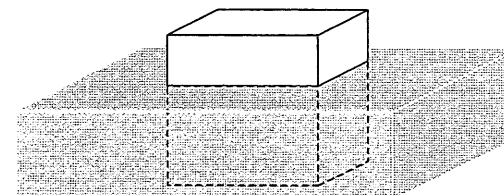


Fig. 17.1

When the block is pushed downwards and then released, it undergoes damped harmonic motion.

- i. Explain why the motion of the block undergoes damping. [1]
- ii. Sketch a labelled graph showing the variation of displacement x from its equilibrium position with time t for a time interval over three periods. [2]
- b. Surface water waves from a constant amplitude wave generator are incident on the block. These causes forced oscillations in the motion of the block. The frequency of the wave generator is varied and resonance was observed at a particular frequency.
- i. Explain the terms *forced oscillations* and *resonance* with reference to the motion of the block. [2]
- ii. Resonance occurs when the water waves incident on the block has a speed of 0.90 m s^{-1} and wavelength 0.30 m .

1. Calculate the frequency of the water waves. [1]

2. Given that the expression for the natural frequency of the oscillating block is given by

$$f = \frac{1}{2\pi} \sqrt{\frac{28}{m}}$$

where f is in Hz and m in kg, calculate the mass of the block. [1]

3. Describe and explain what happens to the amplitude of vertical oscillations of the block after the block has absorbed some water. [2]

[Total: 9]

18. [YJC/H2/Prelims 2018/P2/4 Modified]

Rod P is attached vertically to a horizontal turntable, 0.080 m from the centre Q, as shown in Fig. 18.1.

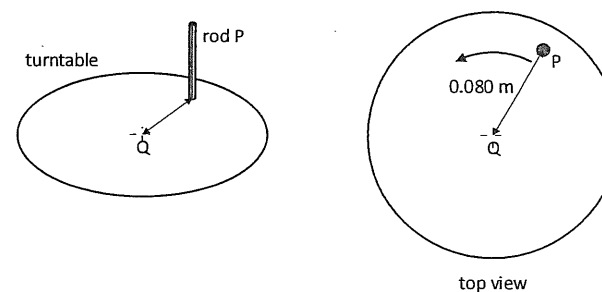


Fig. 18.1

- a. The turntable rotates around Q at 45 rotations per minute. Calculate
- the angular velocity of rod P, [2]
 - the acceleration of rod P and state its direction. [3]
- b. When rod P is illuminated from the side, its shadow on a screen oscillates about O.

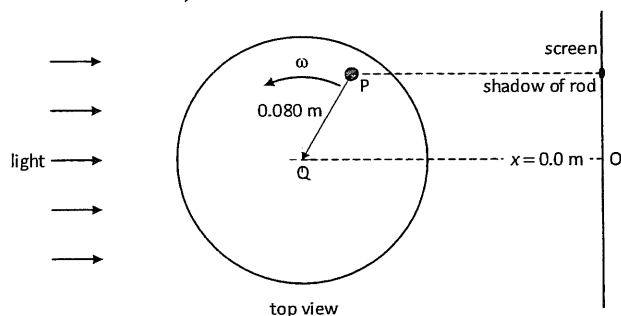


Fig. 17.2

The displacement x_p in metres of the shadow from O may be written as

$$x_p = A \sin(\omega t)$$

where t is time in seconds, A is the amplitude of oscillation and ω is the angular velocity of the rod.

- Using the values given in a.,
 - calculate the period of oscillation of the shadow, and [1]
 - derive an expression for the acceleration a_p of the shadow in terms of t . [2]
- Determine the acceleration of the shadow at $t = 0.20$ s. [2]
- Sketch in Fig. 17.3, with appropriate values, the variation with time t of the acceleration a_p of the shadow of rod P for at least two cycles. [3]

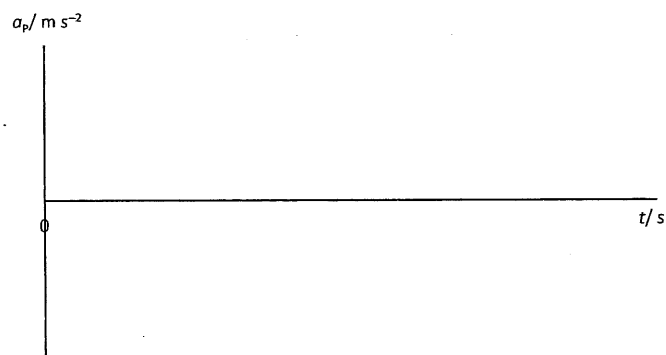
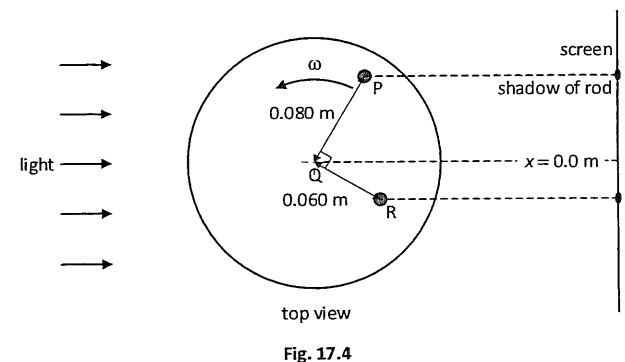


Fig. 17.3

- iv. Fig. 17.4 shows rod R attached to the turntable at a distance of 0.060 m from Q, such that angle PQR is a right angle.



Derive an expression for displacement x_R of the shadow of rod R with the necessary values of its amplitude and angular frequency. [2]

[Total: 15]

